# PHASE 1: UNIVERSAL BILLING, INVENTORY & ACCOUNTING SOFTWARE

## PRODUCT OBJECTIVE

Build a production-quality, modern, scalable **Universal Billing + Inventory + Accounting Software** that can be used by almost any type of business.

This is **Phase 1** of a larger ERP platform.

Phase 1 must focus on:

- Sales billing
- Purchase management
- Inventory
- Product management
- Multi-unit products
- Automatic SKU generation
- Automatic barcode generation
- Customer management
- Supplier management
- Expenses
- Payments and receipts
- Accounts receivable
- Accounts payable
- Cash and bank management
- GST/tax management
- Profit & Loss
- Basic balance sheet
- Trial balance
- General ledger
- Stock valuation
- Business dashboard
- Comprehensive reports
- User roles and permissions
- Audit trail
- Settings
- Data export

DO NOT build this as a toy POS application.

It should feel like a professional business ERP product while keeping Phase 1 focused on billing, inventory and finance.

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# 1. TECHNOLOGY REQUIREMENTS

Use:

- React
- TypeScript
- Vite
- Tailwind CSS
- Supabase
- PostgreSQL
- Supabase Auth
- Supabase Storage
- Row Level Security
- Responsive web application

Use a clean modular architecture so Phase 2 can later add:

- HRMS
- Payroll
- CRM
- Manufacturing
- Projects
- Service management
- Logistics
- Advanced AI
- Mobile application
- E-commerce integrations

Do NOT use WordPress.

Do NOT create a static mockup.

All major screens must be connected to real database operations.

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# 2. APPLICATION STRUCTURE

Create the following main navigation:

1. Dashboard

2. Sales
3. Purchases
4. Products
5. Inventory
6. Customers
7. Suppliers
8. Expenses
9. Payments
10. Accounting
11. Reports
12. Barcode
13. Taxes
14. Users & Roles
15. Settings

Use a professional ERP sidebar.

The sidebar must support:

- Collapse/expand
- Search
- Icons
- Active state
- Nested menus
- Mobile navigation

---

# 3. DASHBOARD

Create a powerful business dashboard.

Top summary cards:

### Sales Today

₹0

### Purchases Today

₹0

### Expenses Today

₹0

### Gross Profit

₹0

## Net Profit

₹0

## Receivables

₹0

## Payables

₹0

## Stock Value

₹0

Add date filter:

- Today
- Yesterday
- This Week
- This Month
- Last Month
- This Quarter
- This Year
- Custom Range

Dashboard charts:

## Sales Overview

Line/bar chart showing:

- Sales
- Purchases
- Expenses
- Profit

## Sales by Category

Pie/donut chart.

## Top Selling Products

Table:

Product | Qty Sold | Revenue | Profit

### Top Customers

Customer | Orders | Sales | Outstanding

### Low Stock

Product | Current Stock | Reorder Level | Status

### Recent Transactions

Show:

- Invoice
- Purchase
- Expense
- Payment
- Receipt
- Stock movement

### Outstanding Summary

- Customer receivables
- Supplier payables
- Overdue amount

---

# 4. COMPANY SETUP

During first login, show setup wizard.

Fields:

- Business name
- Logo
- Business type
- Address
- City
- State
- Country
- PIN/ZIP
- Phone
- Email
- Website
- GSTIN
- PAN
- Tax registration

- Currency
- Financial year
- Invoice prefix
- Invoice starting number

Business type options:

- Retail
- Wholesale
- Manufacturing
- Trading
- Distribution
- Services
- Restaurant
- Pharmacy
- Textile
- Construction
- Agriculture
- Other

Do not change the core system based on industry.

The industry selection should only help with defaults.

---

# 5. PRODUCT MASTER

This is one of the most important modules.

Create a comprehensive Product Master.

Fields:

- Product name
- Product code
- SKU
- Barcode
- Category
- Subcategory
- Brand
- Description
- HSN/SAC
- Tax rate
- Purchase price
- Selling price
- Wholesale price

- Retail price
- MRP
- Minimum selling price
- Reorder level
- Opening stock
- Opening stock value
- Primary unit
- Secondary units
- Conversion factor
- Warehouse
- Rack
- Batch tracking
- Serial tracking
- Expiry tracking
- Active/inactive

---

# 6. UNIVERSAL UNIT SYSTEM

The product system MUST support any measurement unit.

Built-in units:

- Piece
- Kg
- Gram
- Ton
- Quintal
- Litre
- Millilitre
- Meter
- Centimeter
- Feet
- Inch
- Square Feet
- Square Meter
- Cubic Feet
- Cubic Meter
- Box
- Carton
- Pack
- Bundle
- Pair
- Set
- Roll
- Bag

- Bottle
- Can
- Dozen
- Custom

Allow custom units.

Example:

Product:

STEEL ROD

Primary unit:

Kg

Secondary unit:

Meter

Conversion:

1 Meter = 0.75 Kg

Another example:

CEMENT

1 Bag = 50 Kg

Another:

BOTTLE

1 Box = 24 Pieces

Another:

CLOTH

1 Roll = 50 Meter

The system must automatically calculate inventory when different units are used.

---

# 7. PRODUCT PRICING ENGINE

Each product can have multiple prices:

- Purchase price
- Cost price
- Retail price
- Wholesale price
- Dealer price
- Special customer price
- MRP

Allow discount:

- Percentage
- Fixed amount

Support:

- Product discount
- Invoice discount
- Customer discount
- Category discount

---

# 8. AUTOMATIC SKU GENERATION

Provide automatic SKU generation.

Example:

Category:

ELECTRONICS

Product:

Samsung Charger

Automatically create:

ELE-00001

Allow configuration:

PREFIX + CATEGORY + NUMBER

Example:

ELE-CHG-00001

Users can also manually override the SKU.

SKU must be unique.

---

# 9. BARCODE SYSTEM

Automatically generate barcode when creating a product.

Support:

- Code 128
- EAN-13
- EAN-8
- UPC
- QR Code
- Internal barcode

Allow manual barcode entry.

Allow barcode scanning.

Barcode should open the product immediately.

Barcode label printing must support:

- Product name
- SKU
- Barcode
- Price
- MRP
- Unit
- Weight
- Batch
- Expiry
- Company logo

Create a barcode label designer.

---

# 10. WEIGHT / VARIABLE QUANTITY BARCODE

Support variable-weight products.

Example:

Product:

Rice

Barcode:

Contains product + weight.

Scan:

Rice - 12.75 Kg

The billing system should automatically add:

12.75 Kg × price per Kg.

Also support:

- Variable weight
- Variable length
- Variable quantity

---

# 11. CUSTOMER MODULE

Customer fields:

- Customer name
- Customer type
- Phone
- Email
- Address
- GSTIN
- PAN
- State
- Country
- Credit limit
- Credit days
- Opening balance
- Customer price list
- Notes

Customer types:

- Retail

- Wholesale
- Corporate
- Dealer
- Distributor
- Other

Customer profile should show:

- Total purchases
- Total sales
- Outstanding
- Paid
- Pending
- Credit limit
- Last transaction
- Invoice history
- Payment history
- Ledger
- Returns

---

# 12. SUPPLIER MODULE

Fields:

- Supplier name
- Company
- Phone
- Email
- Address
- GSTIN
- PAN
- Credit days
- Opening balance
- Bank details

Supplier profile:

- Total purchases
- Paid
- Outstanding
- Purchase history
- Payments
- Returns
- Ledger

---

# 13. SALES MODULE

Create:

### Sales Invoice

Fields:

- Customer
- Invoice number
- Date
- Due date
- Product
- Barcode
- Quantity
- Unit
- Rate
- Discount
- Tax
- Total

Support:

- Cash sale
- Credit sale
- Partial payment
- Full payment
- Multiple payment methods

Payment methods:

- Cash
- UPI
- Card
- Bank transfer
- Cheque
- Other

---

# 14. FAST BILLING SCREEN

Create a dedicated fast billing/POS screen.

Features:

- Search product
- Barcode scanner
- Product categories
- Product grid
- Keyboard shortcuts
- Quantity
- Unit selection
- Discount
- Tax
- Customer selection
- Payment
- Print
- Save
- Hold invoice
- Resume invoice

The billing screen should be extremely fast.

Allow:

SCAN → ADD → PAYMENT → PRINT

---

# 15. SALES DOCUMENTS

Support:

- Quotation
- Estimate
- Sales Order
- Proforma Invoice
- Tax Invoice
- Delivery Challan
- Credit Note
- Debit Note
- Sales Return

Every document should have:

- Number
- Date
- Customer
- Items
- Tax

- Discount
- Total
- Payment status
- Notes
- Terms

---

# 16. SALES RETURN

Allow full or partial return.

Example:

Invoice:

100 pieces

Customer returns:

20 pieces

System automatically:

- Reduce sales
- Increase stock
- Reverse applicable tax
- Update customer balance
- Update profit

---

# 17. PURCHASE MODULE

Support:

- Purchase quotation
- Purchase order
- Purchase receipt
- Purchase invoice
- Purchase return

Purchase fields:

- Supplier
- Invoice number
- Product

- Quantity
- Unit
- Purchase rate
- Discount
- Tax
- Freight
- Other charges
- Total

---

# 18. PURCHASE RETURN

Allow full or partial returns.

Automatically:

- Reduce inventory
- Reverse purchase
- Update supplier payable
- Update tax
- Update accounting

---

# 19. INVENTORY MODULE

Inventory dashboard:

- Total stock
- Stock value
- Low stock
- Out of stock
- Overstock
- Expiring stock

Stock transactions:

- Opening stock
- Purchase
- Sale
- Sales return
- Purchase return
- Stock adjustment
- Stock transfer
- Stock consumption

---

# 20. MULTI-WAREHOUSE

Support:

- Main warehouse
- Branch warehouse
- Godown
- Store
- Other locations

Stock transfer:

Warehouse A → Warehouse B

Create transfer document.

Track:

- Requested
- Approved
- Sent
- Received

---

# 21. BATCH MANAGEMENT

Optional per product.

Batch fields:

- Batch number
- Manufacturing date
- Expiry date
- Purchase date
- Cost
- Quantity

Alerts:

- Expiring soon
- Expired
- Low stock

---

# 22. SERIAL NUMBER MANAGEMENT

For products such as:

- Electronics
- Machines
- Computers
- Equipment

Track:

Product → Serial Number → Customer → Invoice → Warranty

---

# 23. EXPENSE MANAGEMENT

Create a complete expense module.

Categories:

- Rent
- Electricity
- Internet
- Telephone
- Salary
- Transport
- Fuel
- Maintenance
- Marketing
- Advertising
- Office supplies
- Travel
- Food
- Bank charges
- Insurance
- Professional fees
- Other

Expense fields:

- Date
- Category
- Amount
- Tax
- Vendor
- Payment account

- Description
- Attachment
- Cost centre

Allow receipt upload.

---

# 24. CASH MANAGEMENT

Create:

### Cash Accounts

- Cash
- Petty Cash
- Main Cash
- Branch Cash

Transactions:

- Cash received
- Cash paid
- Cash transfer
- Opening cash
- Closing cash

Show cash balance.

---

# 25. BANK MANAGEMENT

Create bank accounts.

Fields:

- Bank name
- Account name
- Account number
- IFSC
- Opening balance

Transactions:

- Deposit

- Withdrawal
- Transfer
- Payment
- Receipt

Support bank reconciliation structure.

---

# 26. PAYMENT MANAGEMENT

Create unified payment system.

### Receive Payment

Customer → Company

### Make Payment

Company → Supplier

Payment fields:

- Party
- Amount
- Date
- Payment method
- Account
- Reference
- Notes

Automatically update ledger.

---

# 27. RECEIVABLES

Create Accounts Receivable dashboard.

Show:

- Total receivable
- Current
- 1–30 days
- 31–60 days
- 61–90 days

- 90+ days

Customer-wise outstanding.

Invoice-wise outstanding.

Add payment button.

Add reminder button.

---

# 28. PAYABLES

Create Accounts Payable dashboard.

Show:

- Total payable
- Current
- 1–30 days
- 31–60 days
- 61–90 days
- 90+ days

Supplier-wise outstanding.

Invoice-wise outstanding.

---

# 29. ACCOUNTING ENGINE

Build a proper double-entry accounting foundation.

Every financial transaction should create accounting entries.

Example:

Sales invoice:

Debit:
Customer

Credit:
Sales

Credit:
GST payable

Purchase:

Debit:
Purchase/Inventory

Debit:
Input GST

Credit:
Supplier

Expense:

Debit:
Expense account

Credit:
Cash/Bank

Payment:

Debit:
Supplier

Credit:
Bank

Receipt:

Debit:
Bank/Cash

Credit:
Customer

---

# 30. CHART OF ACCOUNTS

Create default chart of accounts.

### Assets

- Cash
- Bank
- Accounts Receivable

- Inventory
- Fixed Assets
- Other Assets

### Liabilities

- Accounts Payable
- GST Payable
- Loans
- Other Liabilities

### Income

- Sales
- Service Income
- Other Income

### Expenses

- Purchases
- Rent
- Electricity
- Salary
- Transport
- Marketing
- Bank Charges
- Other Expenses

Allow users to add custom accounts.

---

# 31. GENERAL LEDGER

Create a complete ledger system.

Filters:

- Account
- Date
- Customer
- Supplier
- Transaction type

Columns:

Date | Particulars | Debit | Credit | Balance

---

# 32. TRIAL BALANCE

Create Trial Balance.

Columns:

Account | Debit | Credit

Show:

- Opening
- Current period
- Closing

Debit total must equal credit total.

---

# 33. PROFIT & LOSS

This is a major Phase 1 feature.

Create a professional Profit & Loss statement.

### Revenue

Sales

Less:

Sales returns

Net Sales

### Cost of Goods Sold

Opening Stock

-

Purchases

-

Direct Expenses

-

Purchase Returns

-

Closing Stock

=

COGS

## Gross Profit

Net Sales - COGS

## Operating Expenses

- Rent
- Salaries
- Electricity
- Marketing
- Transport
- Telephone
- Internet
- Maintenance
- Bank Charges
- Other Expenses

## Net Profit

Gross Profit - Operating Expenses

Show:

- Amount
- Percentage
- Previous period comparison
- Growth %

Allow:

- Monthly
- Quarterly
- Yearly
- Custom date range

---

# 34. BALANCE SHEET

Create basic Balance Sheet.

### Assets

- Cash
- Bank
- Receivables
- Inventory
- Fixed Assets
- Other Assets

### Liabilities

- Payables
- Tax liabilities
- Loans
- Other liabilities

### Equity

- Capital
- Retained earnings
- Current profit

Ensure accounting equation:

ASSETS = LIABILITIES + EQUITY

---

# 35. CASH FLOW

Create basic cash flow report.

Categories:

- Operating
- Investing
- Financing

Show:

Opening Cash

-

Cash Inflows

-

Cash Outflows

=

Closing Cash

---

# 36. STOCK VALUATION

Support configurable valuation method:

- FIFO
- Weighted Average

Show:

- Opening stock
- Purchases
- Sales
- Closing stock
- Stock value
- COGS

---

# 37. PROFIT CALCULATION

For every sales invoice calculate:

Selling price

-

Cost price

=

Gross profit

Show:

- Product profit

- Invoice profit
- Customer profit
- Category profit
- Salesperson profit

Example:

Cost = ₹700

Sale = ₹1,000

Gross profit = ₹300

Gross margin = 30%

---

# 38. TAX / GST MODULE

Create configurable tax engine.

Support:

- GST
- CGST
- SGST
- IGST
- UTGST
- Tax exempt
- Zero rated
- Other custom taxes

Product-level tax.

Invoice-level tax.

Tax reports.

Fields:

- HSN/SAC
- Tax rate
- Tax type

Keep architecture ready for future e-invoice and e-way bill integration.

---

# 39. DISCOUNTS

Support:

- Item discount
- Percentage discount
- Fixed discount
- Invoice discount
- Customer-specific discount
- Quantity discount

Example:

1–10 pieces → ₹100

11–50 → ₹95

51+ → ₹90

---

# 40. CREDIT LIMIT

For every customer:

Credit limit:

₹1,00,000

Current outstanding:

₹85,000

New invoice:

₹30,000

System should warn:

"Credit limit exceeded by ₹15,000."

Allow configurable behaviour:

- Warning
- Block invoice
- Manager approval

---

# 41. DOCUMENT NUMBERING

Configurable prefixes:

INV-000001

PUR-000001

EST-000001

PAY-000001

REC-000001

EXP-000001

Allow:

- Prefix
- Starting number
- Financial-year reset

---

# 42. INVOICE PDF

Generate professional invoice PDFs.

Include:

- Company logo
- Company details
- GSTIN
- Invoice number
- Date
- Customer
- Items
- Quantity
- Unit
- Rate
- Discount
- Tax
- Total
- Amount in words
- Payment details
- Terms
- Signature

Allow print and PDF download.

---

# 43. INVOICE CUSTOMIZATION

Settings:

- Logo
- Header
- Footer
- Terms
- Signature
- Bank details
- UPI QR
- Colour
- Paper size

Support:

- A4
- Thermal
- POS receipt

---

# 44. REPORTING MODULE

Create a dedicated Reports section.

### Sales Reports

- Sales summary
- Sales by product
- Sales by category
- Sales by customer
- Sales by salesperson
- Sales by branch
- Sales by date
- Sales return
- Discount report
- Tax report

### Purchase Reports

- Purchase summary
- Purchase by supplier
- Purchase by product
- Purchase return
- Purchase price history

### Inventory Reports

- Stock summary
- Stock movement
- Stock valuation
- Low stock
- Dead stock
- Fast-moving products
- Slow-moving products
- Batch report
- Expiry report
- Serial number report

### Financial Reports

- Profit & Loss
- Balance Sheet
- Trial Balance
- General Ledger
- Cash Book
- Bank Book
- Cash Flow
- Receivables
- Payables
- Expense report

---

# 45. EXPORT

Every major report should support:

- CSV
- Excel
- PDF
- Print

Filters must be preserved during export.

---

# 46. SEARCH

Create global search.

Search:

- Product
- Customer
- Supplier
- Invoice
- Purchase
- Payment
- Expense
- Barcode
- SKU

Keyboard shortcut:

CTRL + K

---

# 47. NOTIFICATIONS

Notifications for:

- Low stock
- Out of stock
- Expiring batch
- Overdue invoice
- Credit limit exceeded
- Payment received
- Payment due
- Purchase due
- Negative stock
- Failed transaction

---

# 48. USER ROLES

Create role-based permissions.

Default roles:

### Owner

Full access.

### Admin

Full operational access.

### Accountant

Accounting + reports + payments.

### Salesperson

Sales + customers.

### Purchase Manager

Purchases + suppliers.

### Warehouse Manager

Inventory + stock.

### Cashier

Billing + payments.

Allow custom roles.

Permission levels:

- View
- Create
- Edit
- Delete
- Approve
- Export

---

# 49. AUDIT LOG

Record:

- User
- Action

- Module
- Record
- Old value
- New value
- Date
- Time

Examples:

"Invoice INV-001 edited"

"Product price changed"

"Stock adjusted"

"Payment deleted"

"Customer credit limit changed"

---

# 50. SETTINGS

Settings sections:

### Business

Company information.

### Invoice

Numbering and layout.

### Tax

GST/tax settings.

### Products

SKU/barcode settings.

### Units

Unit configuration.

### Inventory

Stock settings.

### Accounting

Chart of accounts.

### Payments

Payment methods.

### Users

Users and permissions.

### Notifications

Alerts.

### Backup

Data export/backup options.

---

# 51. DATABASE DESIGN

Use normalized PostgreSQL tables.

Core tables should include:

companies

branches

warehouses

users

roles

permissions

customers

suppliers

products

product_categories

product_brands

product_units

unit_conversions

product_prices

product_barcodes

product_batches

product_serial_numbers

stock_transactions

stock_balances

sales

sales_items

sales_returns

sales_return_items

quotations

quotation_items

purchase_orders

purchase_order_items

purchases

purchase_items

purchase_returns

purchase_return_items

expenses

expense_categories

payments

payment_allocations

accounts

journal_entries

journal_entry_lines

taxes

tax_rates

invoice_sequences

document_settings

audit_logs

notifications

company_settings

---

# 52. DATABASE ACCOUNTING RULE

DO NOT calculate accounting only on the frontend.

Accounting calculations must be stored through proper backend/database transactions.

Every transaction should be traceable.

For example:

Invoice #INV-001

must connect to:

Sales transaction

Stock transaction

Customer ledger

Tax transaction

Accounting journal

Payment

Profit calculation

This ensures the system remains financially consistent.

---

# 53. STOCK CONSISTENCY

Never allow frontend-only stock calculation.

Every stock movement must create a stock transaction.

Example:

Purchase:

+100 Kg

Sale:

-25 Kg

Return:

+5 Kg

Adjustment:

-2 Kg

Current:

78 Kg

Maintain complete stock history.

---

# 54. NEGATIVE STOCK

Configurable:

Option 1:

Allow negative stock.

Option 2:

Warn.

Option 3:

Block.

Default:

Warn/block depending on user permission.

---

# 55. RESPONSIVE DESIGN

Desktop is the primary experience.

Also support:

- Tablet
- Mobile

Billing screen should work well on touch screens.

---

# 56. UI/UX DESIGN

Design should look like a premium modern ERP.

Avoid:

- Cheap dashboard styling
- Excessive gradients
- Neon colours
- Gaming UI
- Cartoon icons
- Overly rounded cards
- Unnecessary animations

Use:

- Clean white/light interface
- Strong typography
- Professional tables
- Clear spacing
- Compact ERP controls
- Excellent forms
- Consistent buttons
- Clear status badges

Use icons consistently.

---

# 57. TABLE REQUIREMENTS

All major tables must support:

- Search
- Sort
- Filter
- Column selection
- Pagination
- Export
- Bulk selection
- Bulk actions

Example:

Product table:

SKU | Product | Category | Unit | Stock | Cost | Selling Price | Tax | Status | Actions

---

# 58. FORM REQUIREMENTS

Forms should support:

- Validation
- Required fields
- Inline errors
- Auto-save where appropriate
- Keyboard navigation
- Searchable dropdowns
- Barcode scanning
- Product autocomplete

Do not create unnecessarily long forms.

Use tabs/sections for complex product forms.

---

# 59. SAMPLE DATA

Create demo data so the application is immediately usable.

Create:

10 customers

10 suppliers

30 products

Multiple categories

Multiple units

Sales invoices

Purchase invoices

Expenses

Payments

Stock

Accounting transactions

Use realistic data.

The dashboard should look meaningful immediately.

---

# 60. DEMO BUSINESS

Create a demo business:

"ABC Trading & Distribution"

Products:

- Rice
- Cement
- Steel Rod
- Electrical Cable
- Shirt
- Laptop
- Water Bottle
- Office Chair
- Paint

- Cleaning Liquid

Demonstrate different units:

Rice → Kg

Cement → Bag

Steel → Kg + Meter

Cable → Meter

Shirt → Piece

Water → Bottle

Chair → Piece

Paint → Litre

This proves that the ERP works across different industries.

---

# 61. IMPORTANT BILLING WORKFLOW

The following workflow MUST work end-to-end:

Create product

↓

Generate SKU

↓

Generate barcode

↓

Set unit

↓

Set purchase price

↓

Set selling price

↓

Set tax

↓

Add opening stock

↓

Create customer

↓

Create sales invoice

↓

Scan/search product

↓

Enter quantity/unit

↓

Calculate price

↓

Calculate discount

↓

Calculate tax

↓

Calculate total

↓

Receive payment

↓

Reduce stock

↓

Create accounting journal

↓

Update customer ledger

↓

Calculate profit

↓

Update dashboard

↓

Generate invoice PDF

This entire flow must work with real data.

---

# 62. PURCHASE WORKFLOW

Supplier

↓

Purchase Invoice

↓

Product quantity

↓

Unit conversion

↓

Tax

↓

Payment

↓

Stock increases

↓

Supplier payable updates

↓

Accounting updates

↓

Profit/stock valuation updates

---

# 63. EXPENSE WORKFLOW

Expense

↓

Expense category

↓

Amount

↓

Payment account

↓

Accounting entry

↓

Cash/bank decreases

↓

P&L expense increases

↓

Net profit recalculates

---

# 64. RETURN WORKFLOW

Sales return:

Invoice

↓

Select items

↓

Return quantity

↓

Stock increases

↓

Sales decreases

↓

Tax reverses

↓

Customer balance updates

↓

Profit updates

---

# 65. FINANCIAL PERIOD

Support:

- Financial year
- Opening balances
- Closing balances
- Date locking

Allow accountant/admin to lock a period so old transactions cannot be changed without authorization.

---

# 66. DATA INTEGRITY

Implement:

- Foreign keys
- Unique constraints
- Check constraints
- Database transactions
- RLS policies
- Proper indexes
- Audit logs

Do not rely on frontend validation alone.

---

# 67. PERFORMANCE

The application should be designed to support:

- Thousands of products
- Thousands of customers
- Thousands of invoices
- Large stock transaction history

Use:

- Pagination
- Database indexes
- Server-side filtering
- Efficient queries
- Lazy loading where appropriate

Do not load the entire database into the browser.

---

# 68. PHASE 1 BOUNDARY

Do NOT implement advanced Phase 2 modules yet:

- Full HRMS
- Payroll
- Recruitment
- CRM automation

- Manufacturing MRP
- Production planning
- Advanced project management
- Advanced logistics
- Advanced AI automation
- E-commerce marketplace integrations

However, the database and architecture must be designed so these modules can be added later without rebuilding the core.

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# 69. FUTURE PHASE 2 READY

Prepare architecture for:

### HRMS

Employee → Attendance → Leave → Salary → Payroll

### CRM

Lead → Opportunity → Quotation → Sale

### Manufacturing

BOM → Production → QC → Finished Goods

### AI

Document → AI Extraction → Validation → ERP Transaction

### Logistics

Order → Picking → Packing → Dispatch → Delivery

### Service

Ticket → Technician → Job Card → Invoice

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# 70. FINAL QUALITY REQUIREMENT

Do not simply create screens.

Every important button must perform a real operation.

Do not create fake dashboard numbers.

Do not use hardcoded financial calculations.

Do not create disconnected modules.

The system must behave like one integrated ERP.

The most important principle is:

**ONE TRANSACTION → AUTOMATICALLY UPDATE EVERYTHING RELATED TO IT.**

For example:

A ₹10,000 sales invoice should automatically update:

- Sales
- Customer outstanding
- Inventory
- Tax
- Cost of goods sold
- Gross profit
- Net profit
- General ledger
- Trial balance
- Cash/bank if paid
- Dashboard
- Reports
- Audit trail

Similarly, every purchase, expense, payment, receipt and return must flow through the appropriate modules.

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# PHASE 1 SUCCESS CRITERIA

The finished application should allow a real small/medium business to operate its core financial and billing activities without needing another basic billing application.

A user should be able to:

1. Create company
2. Create products
3. Configure any unit
4. Create unit conversions

5. Generate SKU
6. Generate barcode
7. Print barcode
8. Create customers
9. Create suppliers
10. Create purchase
11. Receive inventory
12. Sell products
13. Scan barcode
14. Bill by piece
15. Bill by Kg
16. Bill by gram
17. Bill by meter
18. Bill by litre
19. Bill by box
20. Bill by custom unit
21. Apply discounts
22. Calculate GST/tax
23. Receive payments
24. Make supplier payments
25. Record expenses
26. Track receivables
27. Track payables
28. Track stock
29. Track batches
30. Track serial numbers
31. Transfer stock
32. Generate invoices
33. Generate reports
34. View customer ledger
35. View supplier ledger
36. View general ledger
37. View trial balance
38. View Profit & Loss
39. View Balance Sheet
40. View Cash Flow
41. Calculate product profit
42. Calculate invoice profit
43. Calculate stock value
44. Export reports
45. Manage users
46. Track audit history

The result should be a **professional Universal Billing + Inventory + Accounting ERP foundation**, ready for Phase 2 expansion into a complete ERP platform.